

Exploring the Reasons behind Attrition: A Power BI Report

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Research Question

1. What are key factors contributing to employee attrition?

Introduction

The goal of this project is to analyze employee attrition and retention (whether employees stay or leave a company). Using a Kaggle dataset, we explore how various employee attributes (such as salary, education, and job satisfaction) impact employee retention. With Power BI, we transformed and visualized the data, allowing for insights into what factors contribute most to attrition and how these patterns can inform a company's human resources and business strategies. Our visualizations and analyses suggest that salary is a significant factor in employee attrition, with employees in lower salary ranges exhibiting higher turnover rates, whereas education, job satisfaction, and job level also play a role albeit to a lesser extent. Employee attrition remains a pressing concern for businesses, as it directly impacts operational costs and efficiency¹. By analyzing factors salary, education, and job satisfaction with Power BI, we aim to provide actionable insights into which attributes have the most impact on employee attrition in order to develop strategies to reduce turnover and improve retention.

Our interest in this subject matter developed from a conversation with a plant manager for the wholesale distributor, Trinidad Benham Corporation. Group member Liam Fisher was speaking with the manager during a site visit earlier this year about his main concerns as he and his team start the 2025 calendar year, and the most pressing item on his mind. Employee retention. He sees high turnover rates amongst his forklift drivers and machine operators and would love to keep his workers employed for longer periods of time.

¹ 1. By: Carol Warner, "The Real Cost of Employee Turnover Now," HRMorning, April 23, 2025, <https://www.hrmorning.com/articles/real-cost-employee-turnover/>.

Dataset Description

The data set used for this project is a synthesized table on Kaggle from IBM's Human Resources Department on a fictional company, covering various employee characteristics such as salary, tenure, education level, various satisfaction metrics, and crucial to this study, attrition.²

- EmployeeID: Unique identifier for each employee.
- Age: Age of the employee.
- Job Satisfaction: A numerical score representing the employee's job satisfaction level.
- MonthlyIncome: Employee's monthly salary.
- Education: Education level (e.g., High School, Bachelor's, Master's, etc.).
- Attrition: Whether the employee left the company (1 for Yes; 0 for No).

Justification: The dataset provides a comprehensive view of employee characteristics and their relationship with attrition, making it suitable for this analysis.

Data Model

- Cleaning and Transformation: We removed unnecessary columns, filled in missing data, and formatted the columns (e.g., converting salary to numerical values).
- Relationships: We created unique primary and foreign keys to connect our dimension tables to our fact table, which contained survey data such as job satisfaction, work-life balance, and attrition count. Transforming a flat table to a multi-dimensional data model, we illustrated

² IBM HR Analytics Employee Attrition & Performance, distributed by Kaggle, <https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

consequential relationships between our education, job roles, salary, and attrition. For simplicity, we mainly structured our model as star-shaped but the secondary connection between DimEmployee and DimYears leans more toward a snowflake schema.

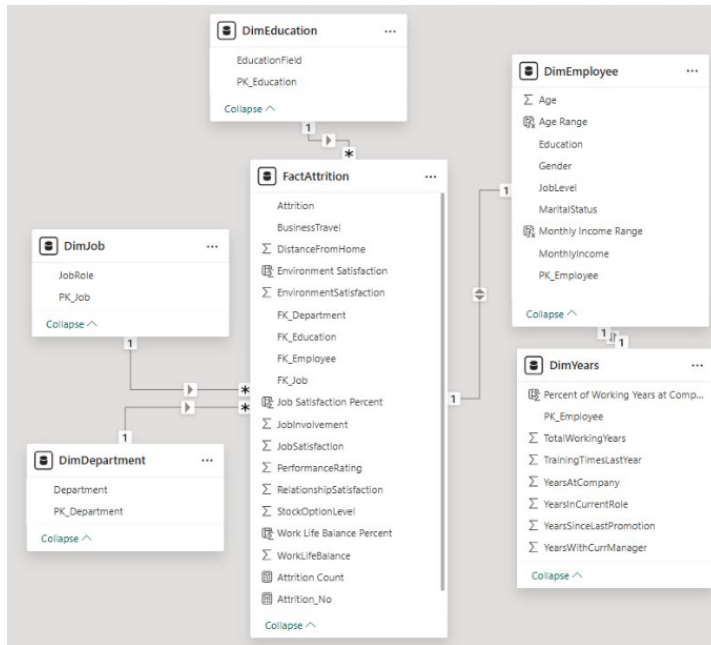


Figure 1. Data Model

- Calculated Columns: We created new columns like Monthly Income Range (as shown below) by using DAX functions divide monthly income into subsets to better visualize our data that is more readable as well as interactive for viewers. Other measures include Age Range, where we split age into digestible categories such as 20s, 30s, 40s, etc. Another important measure was Percent Working Years at Company, so we could measure the proportion of years spent at the company of interest versus elsewhere.

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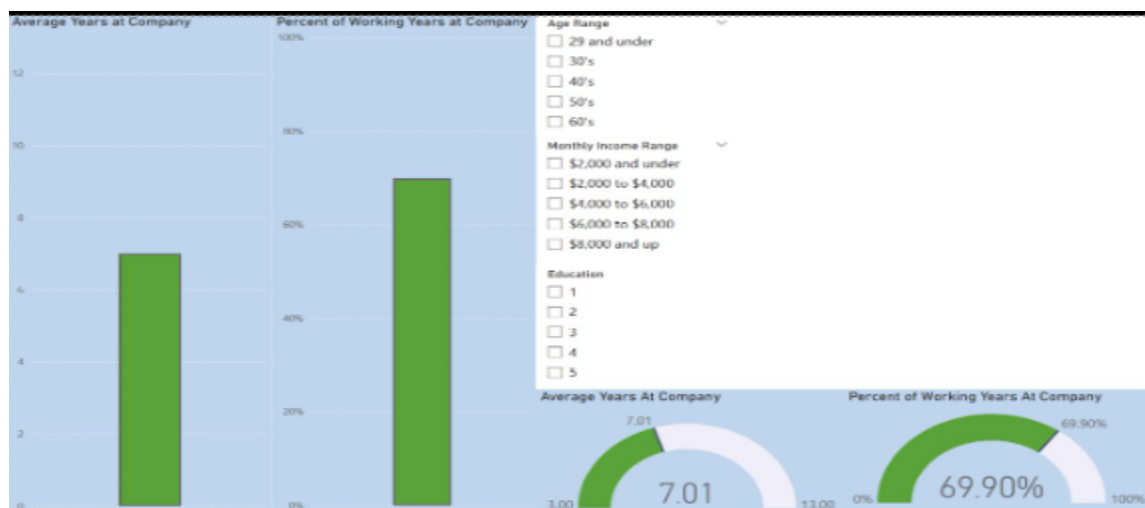
1 Monthly Income Range =
2 SWITCH(
3     TRUE(),
4     'DimEmployee'[MonthlyIncome] <= 2000, "$2,000 and under",
5     'DimEmployee'[MonthlyIncome] > 2000 && 'DimEmployee'[MonthlyIncome] <= 4000, "$2,000 to $4,000",
6     'DimEmployee'[MonthlyIncome] > 4000 && 'DimEmployee'[MonthlyIncome] <= 6000, "$4,000 to $6,000",
7     'DimEmployee'[MonthlyIncome] > 6000 && 'DimEmployee'[MonthlyIncome] <= 8000, "$6,000 to $8,000",
8     'DimEmployee'[MonthlyIncome] > 8000, "$8,000 and up",
9     BLANK()
10 )

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Figure 2. Example of Created Measure

Visualizations

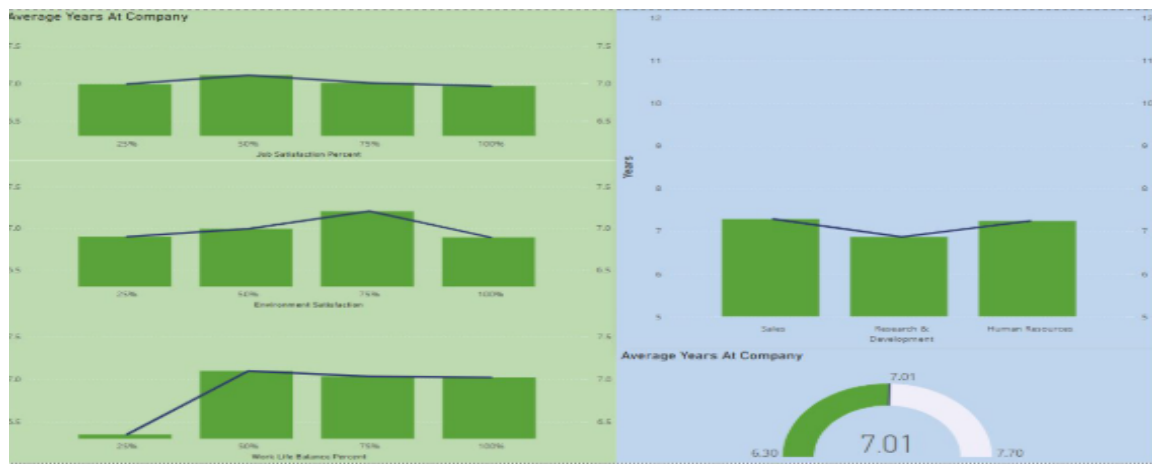
Chart 1: Salary vs. Average Years at Company



- Purpose:** This chart shows how salary influences average years spent at the company as well as the percentage of a person's working years spent at the given company. We used a bar chart, a gauge, and a splice to show how a person's total years spent and the percentage of total working years at the company varies by salary ranges.
- Insight:** The data revealed that employees who fell into higher monthly salary ranges have spent, on average, more years at the company. This suggests that the company might want to focus on increasing salary packages for lower-level employees to improve retention. We also found that as

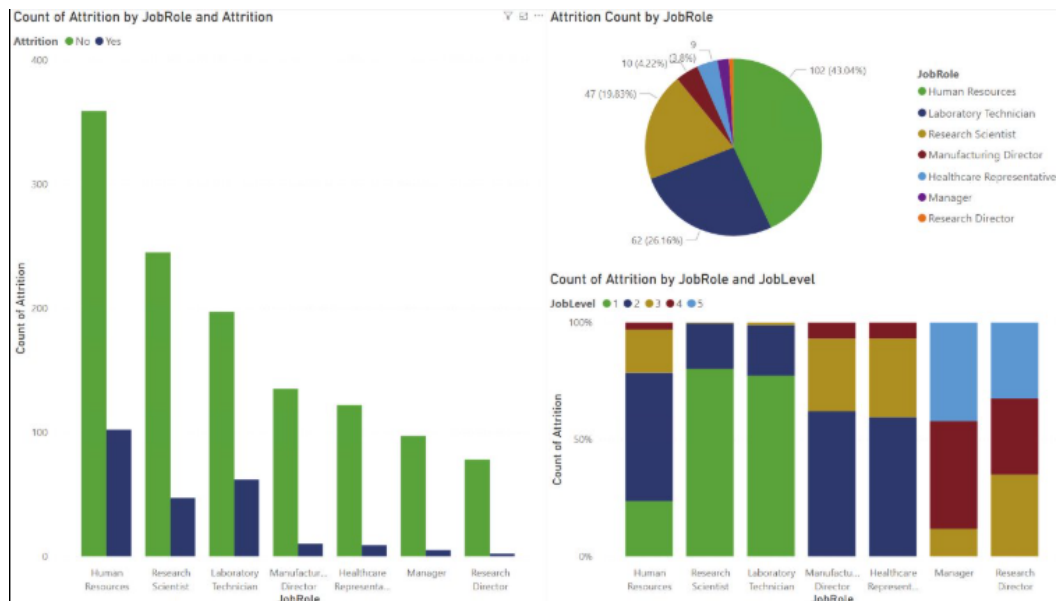
monthly income brackets increase, the percentage of total working years at the company decreases slightly. Initially, we thought these two observations were at odds with one another, but this latter observation can be explained by the fact that higher earners have typically worked more years in total, and so even if they have spent many years at a given company, when viewed as proportional to the rest of their career, this number will still be relatively lower.

Chart 2: Job Satisfaction vs. Attrition



- Purpose: To explore how job satisfaction influences attrition. A line and clustered column chart was used to show the relationship between job satisfaction scores and average years at the company.
- Insight: We expected to find employees with lower job satisfaction spending less time at the company as compared to those who have higher levels of job satisfaction. However, this is not what we observed, as employees who rated their satisfaction as two out of 4 spent the highest average number of years at the company. We then further investigated similar metrics around work environment satisfaction and work life balance satisfaction, but similarly did not find a correlation between higher rankings in these measures and higher average years spent at the company. This inspired us to investigate other factors such as job role, monthly income, and education, as you will see in the diagrams found below.

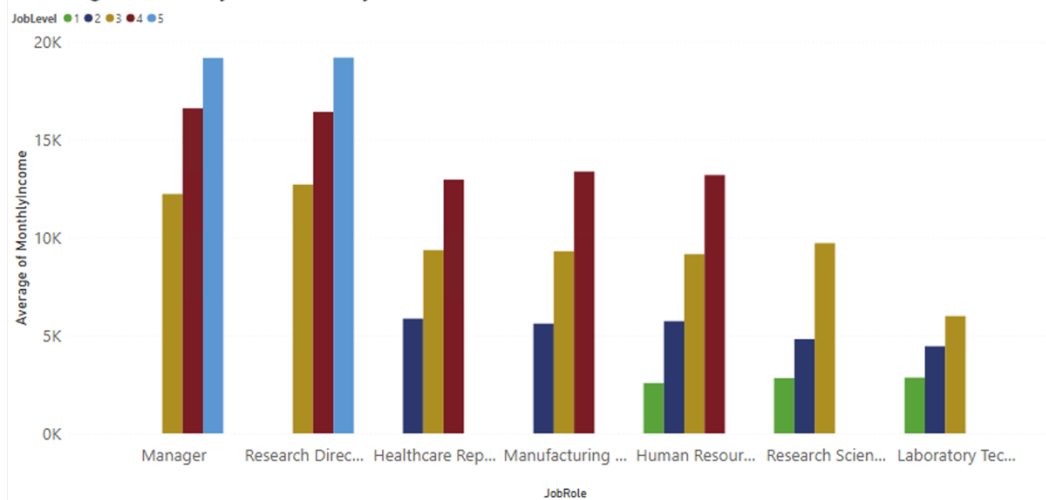
Chart 3: Job Role vs. Attrition



- Purpose: To explore the relationship between job role and attrition and to see in what ways job role may contribute to attrition. Three charts were created for this: first, a clustered column chart to show attrition versus non-attrition for each job role; second, a pie chart to show the proportion of attrition by job role; third, 100% stack column chart to further explore the proportion of attrition by job role but drilled down by job level, which ranges from entry level (1) to an executive level (5).
- Insight: First, we must acknowledge the imbalance in our data set: most people stay with the company at the time of the survey rather than leave when analyzed by job role. We see that human resources have the highest count of attrition and that's further shown in the pie chart where they subsequently have the highest proportion of attrition as well out of any job role. When we drill down by job level, we see that lower-level employees are the ones to commit attrition rather than higher-level. We see a discrepancy in our insights Manager and Research Director only because the nature of those positions does not allow for lower-level positions.

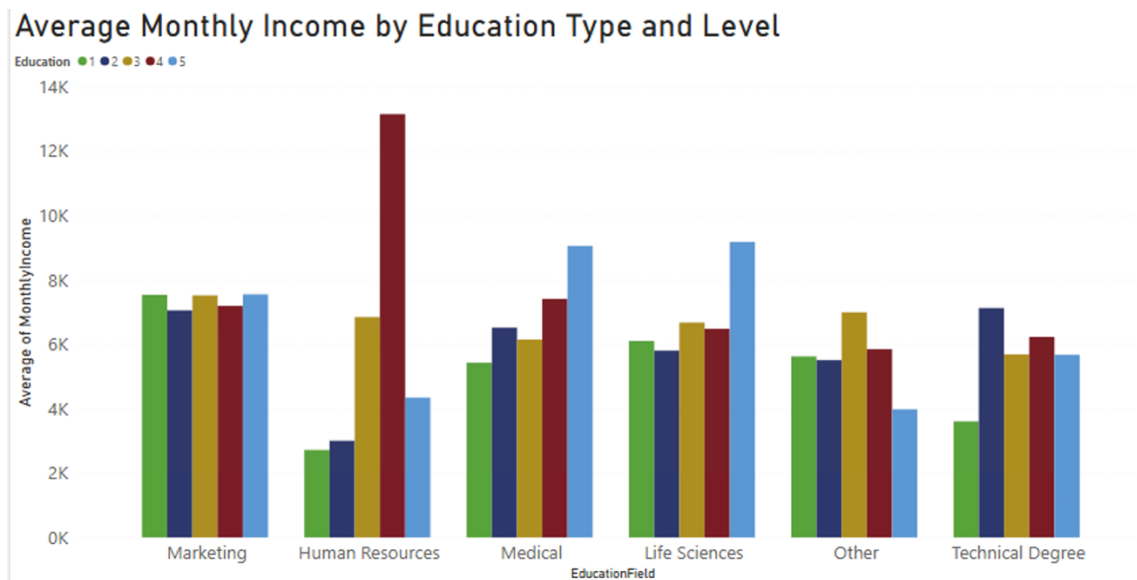
Chart 4: Monthly Income vs. Job Role

Average Monthly Income by Job Role and Level



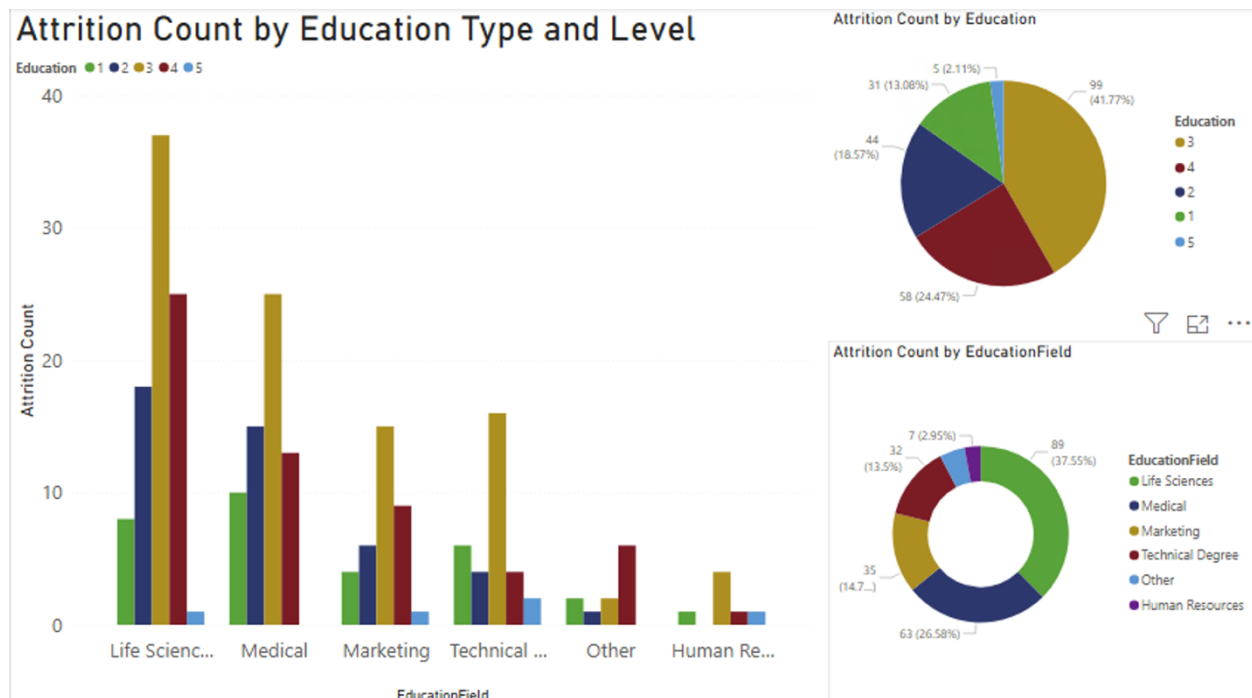
- Purpose: To explore the relationship between average monthly income by job role and job level and to see if this drilled-down relationship will provide insight into how job role influences attrition.
- Insight: We found that average monthly income by job role mirrors our finding above. Lower job levels have lower average monthly income, where we found above that lower job levels for each category of job role had the highest counts of attrition. The three job roles with the highest levels of attrition are the three job levels with the lowest average monthly salaries: Human Resources, Research Scientist, and Laboratory Technician. This further shows us the importance of financial compensation in preventing attrition.

Chart 5: Education Level vs. Attrition



- Purpose: Illustrate the relationship between average monthly salary, education type, and education level—from below high school (1) to PhD (5)—to see if average monthly salary shows the same relationship with attrition except now explored through the lens of education of employees. We used a clustered column chart to break down average monthly salary but education type and level.
- Insight: We found for most education categories monthly pay varied little across education level except instance such as Human Resources, where there was a drastic difference once an employee reached master's level in their education. We also found that there was a steady increase, albeit small, in average monthly as education level increased for the Life Sciences and Medical. Interestingly, we also found that higher education level had lower monthly averages once an employee exceeded high school for technical degrees. We find these insights concerning in that higher education levels are not rewarded with higher salaries, and if they are, most cases show the difference in salary is marginal.

Chart 6: Education Level vs. Attrition



- Purpose: These charts compare employee attrition counts across different education types and levels. We used a stacked column chart to display the attrition counts for employees with various educational backgrounds. A pie chart was used to show the proportion of attrition by education level, whereas a donut chart showed the proportion of attrition by education type.
- Insight: For the most part, employees with higher education, especially bachelor's or master's degrees, had higher turnover, which might suggest that these employees are more likely to seek other opportunities for career advancement. It is also worthy to note that we found above that there were mostly marginal differences in average monthly salary across education levels. This lack of compensation for education level may explain why attrition is higher for more educated employees. For example, Life sciences have the highest level of attrition count out of all the education types and a bachelor's degree has the most counts of attrition for this category, and as we had shown in our previous visual, there were marginal differences in average monthly salary across education levels, which makes us believe that if education levels are not properly

compensated then they are more likely to leave the company to find better pay elsewhere for their degree.

Conclusion

1. **Salary and Years at Company:** The Salary vs. Average Years at Company chart clearly shows that employees with lower monthly salaries spend, on average, less time at the company. This is a crucial finding for the company, as it indicates that increasing salaries for lower-paid employees could significantly reduce turnover.
2. **Job Satisfaction:** Employees with lower job satisfaction levels are more likely to leave the company. The Job Satisfaction vs. Attrition chart reveals that improving job satisfaction, such as offering career development opportunities or improving management practices, could help retain employees.
3. **Salary versus Job Role:** Lower-level employees have higher counts of attrition, most likely due to lower average monthly salaries. We posit that financially compensating lower-level employees would lead to lower turnover.
4. **Education and Attrition:** The Education vs. Attrition chart shows that employees with higher education are more likely to leave the company. This suggests that offering growth opportunities or career advancement programs may help retain these employees, who may otherwise seek better opportunities elsewhere.

The analysis shows that salary, job satisfaction, job role, and education level are all factors influencing attrition of the employees surveyed in this data set. To reduce attrition, the company should consider increasing salaries for lower-paid employees, improving job satisfaction through career development and training, and offering retention incentives for employees with higher educational qualifications. These

findings will be quite useful for the plant manager of Trinidad Benham Corporation as he and his team go about allocating their funds for 2025. Based on our findings, they should consider apportioning more funding to their lower-paid employees to keep them employed with the company longer.

Response to Feedback

The feedback for our group project provides valuable insights into areas where we could improve and expand upon our analysis. The feedback highlights the effectiveness of our presentation, with praise for the clarity of our visualizations and the logical flow from research question to results. Our decision to integrate real-world context through Liam's workplace experience was also well received, adding relevance to our findings. However, several suggestions were made to enhance the depth of our analysis. For example, while our data model included many measures, we could have provided more insight into some of the less discussed queries. Additionally, the feedback noted that expanding our dataset to include data from multiple organizations or industries could have provided a broader perspective on attrition. The feedback also raised thought-provoking questions about the impact of education, industry, and job level on attrition rates, urging us to explore the unexpected higher attrition among bachelor's degree holders and the specific circumstances in industries like life sciences. While the feedback was constructive, we feel it focused on aspects that were not directly addressed in the project's grading rubric. The suggestions, such as exploring statistical analysis for correlation or expanding the scope of our dataset, are indeed valid but may be beyond the project's scope or data limitations. Overall, the feedback provides useful direction for further refining our analysis and would be valuable for future projects that aim to delve deeper into the complexities of employee attrition.

Bibliography

IBM HR Analytics Employee Attrition & Performance, distributed by Kaggle,

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

Warner, By: Carol. "The Real Cost of Employee Turnover Now." HRMorning, April 23, 2025.

<https://www.hrmorning.com/articles/real-cost-employee-turnover/>.